ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

DISTANCE LEARNING PROGRAMME

(Academic Session : 2024 - 2025)

JEE (Advanced)
AIOT
20-04-2025

JEE(Main + Advanced) : LEADER TEST SERIES / JOINT PACKAGE COURSE

Test Type : Full Syllabus

ANSWER KEY — PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	B	B	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,D	A,B,C	B,C	A,B,D	B	A,B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	0.03 to 0.04	12.50	5.60	16.00	4.00	1.25 to 1.27
SECTION-II (ii)	Q.	7	8	9			
	A.	1080	2	4			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	A	C	C	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	C,D	B,C	B,C	A,B,C	B,C,D	A,B,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.00	5.00	28.00	30.00	181.25	8.75
SECTION-II (ii)	Q.	7	8	9			
	A.	81	10	84			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	B	C	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,C	B,C,D	B,C,D	B,C	A,C	A,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	7.00	24.00	26.00	3.00	3.00	0.50
SECTION-II (ii)	Q.	7	8	9			
	A.	32	4	5			

HINT – SHEET

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PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (B,C)

When the switch 'S' is opened,

$\tau = RC = 1.5$ sec

Voltage across the capacitor,

$V_C = 10(1 - e^{-t/1.5})$ for $t < T$

The current, $i = \frac{dQ}{dt} = C\frac{dV_C}{dt}$

$= \frac{200}{3} \times 10^{-6}(e^{-t/1.5})$ for $t < T$

Voltage drop across 100 k μ resister,

$V = iR = \frac{20}{3}e^{-t/1.5}$ for $t < T$

After $t = T$,

$\tau = 100\ k\Omega \times 10\ \mu F = 1$ sec

Voltage across capacitor,

$V_C = V_0 t^{-[(t-T)/\tau]} = 10(1 - e^{-(T/1.5)}), e^{-[(t-T)/1]}$

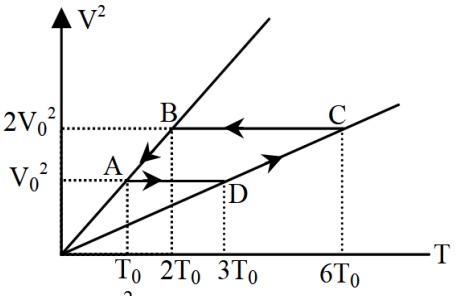
The charge on capacitor,

$Q = V_C \cdot C$

The current through 100 k μ resister

$i = -C\frac{dV_C}{dt} = 10^3\left(1 - e^{-T/1.5}\right)\left(-e^{-(t-T)/t}\right)$

8. Ans (A,B,D)



$B \rightarrow A\ \frac{V^2}{T} = \text{constant}$

$\therefore T_B = 2T_0$

$D \rightarrow C\ \frac{V^2}{T} = \text{constant}$

$\Rightarrow T_C = 6T_0$

$\therefore \frac{T_B + T_C}{2} = 4T_0$

$B \rightarrow A\ \frac{V^2}{T} = \text{constant}$

$\frac{V^2}{\left(\frac{PV}{nR}\right)} = \text{constant}$

$PV^{-1} = \text{constant}$

polytropic index $m = -1$

$C = \frac{R}{1-m} + \frac{Rf}{2}$

$C = 2R$

(same for $D \rightarrow C$)

$D \rightarrow C : Q = \Delta U + W$

$nc\Delta T = nc_v\Delta T + W$

$1 \times 2R \times 3T_0 =$

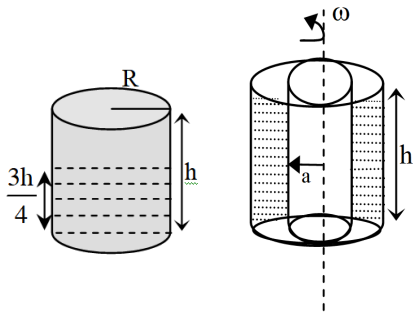
$1 \times \frac{3R}{2} \times 3T_0 + W$

$W = \frac{3RT_0}{2}$

PART-1 : PHYSICS

SECTION-II (ii)

8. Ans (2)



g is absent

$g_{eff} = \omega^2 r \rightarrow$ (r = distance from axis)

volume of liquid = constant

$\frac{3\pi R^2 h}{4} = \pi R^2 h - \pi a^2 h$ $(h = 12cm)$

$a = \frac{R}{2}$

$P_{wall} = P_0 + \rho\int_a^R \omega^2 r\, dr$

$P_{wall} = P_0 + \frac{\rho\omega^2}{2}(R^2 - a^2)$

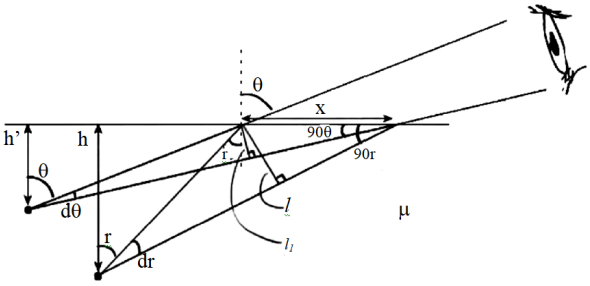
$P_{wall} = P_0 + \frac{3\rho\omega^2 R^2}{8}$

$P_{wall} = 10^4 + \frac{3 \times 800 \times \left[\frac{250}{\sqrt{3}} \times 4 \times 10^{-2}\right]^2}{8}$

$P_{wall} = 2 \times 10^4 Nm^{-2}$

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9. Ans (4)



Rays are marginal. Shift is perpendicular to surface as well as along the surface.

$\ell' = \left(\frac{h'}{\cos\theta}\right) d\theta = x \sin(90^\circ - \theta)$

$\ell = \left(\frac{h}{\cos r}\right) dr = x \sin(90^\circ - r)$

Divide : $\frac{h'}{h}\frac{d\theta}{dr} = \frac{\cos^2\theta}{\cos^2 r} \ldots . (1)$

Also, $\sin\theta = \mu \sin r$

$\Rightarrow \cos d\theta = \mu \cos r\, dr$ ....(2)

From 1,2 : $h' = \frac{h\cos^3\theta}{\mu\cos^3 r} = \frac{h\cos^3\theta}{\mu\left(1 - \frac{\sin^2\theta}{\mu^2}\right)^{3/2}}$

put $h = 8\sqrt{3}$m

$\theta = 60^\circ$

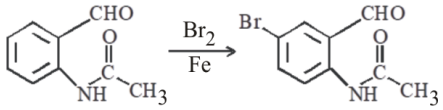
$\mu = \sqrt{3/2}$

h' = 4 m

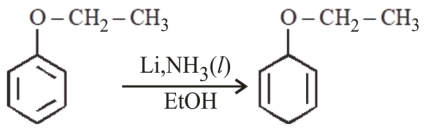
PART-2 : CHEMISTRY

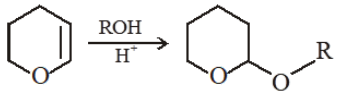
SECTION-I (i)

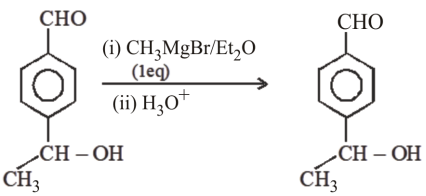
1. Ans (A)

(A) 

More activating so, bromination occurs at its para

(B) 

(C) 

(D) 

2. Ans (C)

$HA + KOH \longrightarrow K^+A^- + H_2O$

At equivalence point,

Number of equivalents of HA

= Number of equivalents of KOH

= Number of equivalents of salt

$= \frac{60}{1000} \times 0.1 \times 1 \Rightarrow 6 \times 10^{-3}$

Now HCl is added to the salt solution

	K^+A^- +	HCl ⟶	HA	+ KCl
t = 0	0.006 moles	0.002 moles		
t = t	0.004 moles	0	0.002 moles	

It is an acidic buffer solution

$p^H = p^{Ka} + \log\frac{[A^-]}{[HA]}$

$p^{Ka} = 5$

Now, p^H will be due to salt hydrolysis :

$K_n = \frac{K_w}{K_a} = \frac{10^{-14}}{10^{-5}} = 10^{-9}$

$10^{-9} = \frac{Ch^2}{1-h}$

Suppose h < < < 1

$10^{-9} = \left(\frac{5\times10^{-3}}{10^{-1}}\right).h^2$

$\frac{1}{5} \times 10^{-7} = h^2$

$2 \times 10^{-8} = h^2$

$h = \sqrt{2} \times 10^{-4}$

$[OH^-] = Ch$

$= (5 \times 10^{-2})\sqrt{2} \times 10^{-4}$

$= 5\sqrt{2} \times 10^{-6}$

$p^{OH} = 6 - \log 5\sqrt{2}$

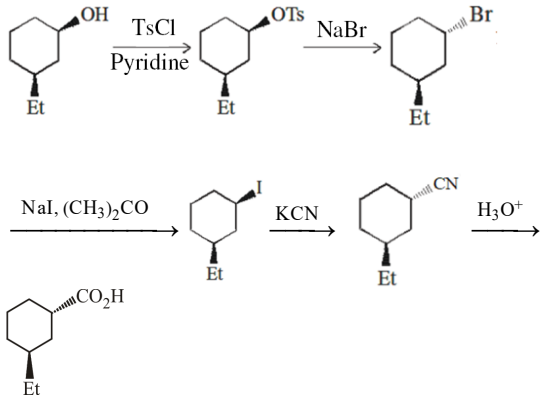
$= 6 - \left\{\log 5 + \frac{1}{2}\log 2\right\}$

= 6 - 0.85 = 5.15

p^H = 14 - 5.15 =8.85

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3. Ans (C)



4. Ans (D)

$[CoF_6]^{3-} \longrightarrow sp^3d^2$ and paramagnetic

$[NiCl_4]^{2-} \longrightarrow sp^3$ and paramagnetic

$[Ni(CO)_4] \longrightarrow sp^3$ and diamagnetic

$[Cu(NH_3)_4]^{2+} \longrightarrow dsp^2$ and paramagnetic

PART-2 : CHEMISTRY

SECTION-I (ii)

6. Ans (B,C)

(A) The distance between two nearest Zn^{+2} ions is $\frac{a}{\sqrt{2}} = 5A°$, then the distance between two nearest S^{2-} ions is $\frac{a}{\sqrt{2}} = 5A°$.

(B) In NaCl (rock salt) structure; for any anion (placed at corner), distance of second nearest cation is $\frac{\sqrt{3}}{2}a$ (present at body centre) and coordination number is 8.

(C) The third nearest anion for any anion placed at corner is present at centre of opposite face at a distance of $\sqrt{\frac{3}{2}}$ a.

(D) The packing fraction of CsCl is

$$P.F. = \frac{\frac{4}{3}\pi(r_+^3 + r_-^3)}{a^3}$$

$$= \frac{\frac{4}{3}\pi(r_+^3 + r_-^3)}{\frac{8}{3\sqrt{3}}(r_+ + r_-)^3}$$

$$= \frac{\sqrt{3}\pi}{2} \times \frac{10.745}{42.875} = \frac{\sqrt{3}\pi}{8}$$

8. Ans (A,B,C)

$$CaCO_3 + 2HCl \longrightarrow CaCl_2 + H_2O + CO_2$$

moles x 2x

$$MgCO_3 + 2HCl \longrightarrow MgCl_2 + H_2O + CO_2$$

moles y 2y

Number of equivalents of HCl added

$$= \frac{250}{1000} \times 0.4 \times 1$$

= 0.1 equivalents

Number of equivalents of HCl neutralized by Na_2CO_3 (in 10ml) =

$$\frac{10}{1000} \times \frac{1}{20} \times 2 = 0.001 \text{ equivalents}$$

Total equivalents of HCl neutralized by Na_2CO_3 in 400 ml = 0.001 × 40 = 0.04 equivalents

No. of moles of HCl used by $CaCO_3$ & $MgCO_3$ are

= 0.1 - 0.04 = 0.06 moles

according to equations

2x + 2y = 0.06

x + y = 0.03 ----------(1)

and

100x + 84y = 2.84 ---------- (2)

Solving (1) & (2)

x = 0.02 moles

y = 0.01 moles

(A) % mass of $CaCO_3 = \frac{2}{2.84} \times 100 = 70.42\%$

(B) % moles of $MgCO_3 = \frac{0.01}{0.03} \times 100 = 33.33\%$

(C) Number of equivalents of HCl neutralised by $Na_2CO_3 = 0.06$

(D) Number of moles of HCl neutralised by Na_2CO_3 is 0.04

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9. **Ans (B,C,D)**

(A) In leaching of aluminium ore, a small amount of freshly prepared sample of hydrated Al_2O_3 is used to cause precipitation (Bayer process)

(B) Wrought iron is purest commercial iron and impurities are oxidised with hematite.

$$Fe_2O_3 + C \xrightarrow{\Delta} Fe + CO_2$$

$$Fe_2O_3 + Si \xrightarrow{\Delta} Fe + SiO_2$$

(C) Copper matte consists of Cu_2S + FeS (very small amount)

(D) Extraction of zinc from ZnS involves roasting followed by carbon reduction.

10. **Ans (A,B,C)**

(A) Colloid-ion is a 4% solution of nitro-cellulose in a mixture of alcohol and ether. (NCERT)

(B) A mixture of milk and water appears blue when viewed by the reflected light and red when viewed by the transmitted light. (NCERT)

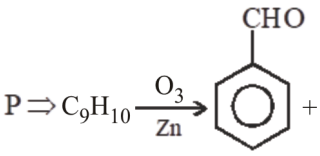
(C) Starch, gum, clay ⟶ Negatively charged sol.
Methylene blue ⟶ Positively charged sol. (NCERT)

(D) Correct

PART-2 : CHEMISTRY

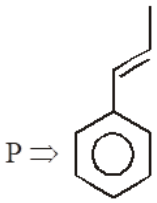
SECTION-II (i)

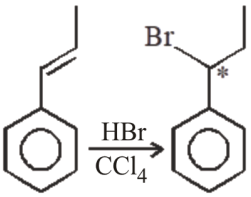
1. **Ans (2.00)**



Positive iodoform test

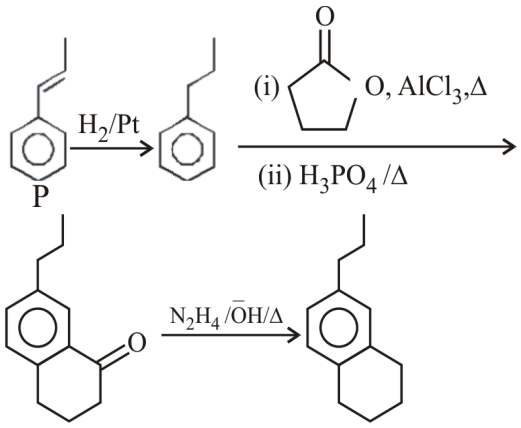
So,





Total products = 2

2. **Ans (5.00)**



3. **Ans (28.00)**

$$K_4[Fe(CN)_6] + Fe^{+3} \rightarrow Fe_4[Fe(CN)_6]_3$$

(A) Deep blue ppt. (D)

$$K_3[Fe(CN)_6] + Cu^{2+} \xrightarrow{\text{Acidic medium}} Cu_3[Fe(CN)_6]_2$$

(B) Green colour ppt. (E)

$$K_3[Fe(CN)_6] + Fe^{+3} \rightarrow Fe[Fe(CN)_6]$$

$$\xrightarrow{SnCl_2 \text{ solution}} Fe_4[Fe(CN)_6]_3$$

(B) Brown coloration Deep blue ppt. (D)

Number of unpaired electron in $K_3[Fe(CN)_6]$ is 'p' = 1

Number of atoms in a molecule of $Cu_3[Fe(CN)_6]_2$ is 'q'= 29

|p - q| = 28

4. **Ans (30.00)**

$$K_4[Fe(CN)_6] + Fe^{+3} \rightarrow Fe_4[Fe(CN)_6]_3$$

(A) Deep blue ppt. (D)

$$K_3[Fe(CN)_6] + Cu^{2+} \xrightarrow{\text{Acidic medium}} Cu_3[Fe(CN)_6]_2$$

(B) Green colour ppt. (E)

$$K_3[Fe(CN)_6] + Fe^{+3} \rightarrow Fe[Fe(CN)_6]$$

(B) Brown coloration

$$\xrightarrow{SnCl_2 \text{ solution}} Fe_4[Fe(CN)_6]_3$$

Deep blue ppt. (D)

$$K_4[Fe(CN)_6] + H_2SO_4(\text{conc.}) \rightarrow CO$$

(x) = 28

oxidation state of central metal in 'A' is y = 2

|x + y| = 30

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5. **Ans (181.25)**

$W_{ABCDA} = W_{AB} + W_{BC} + W_{CD} + W_{DA}$

Along AB

$P \propto V \Rightarrow P = KV$ or $T = K_1V^2 \Rightarrow \frac{T_1}{T_2} = \left(\frac{V_1}{V_2}\right)^2$

At point A; P_0, $4V_0$, $T_A = 100K$

At point B; $2P_0$, $8V_0$, $T_B = 400K$

Along BC, volume is constant

$P \propto T$

At point C; $4P_0$, $8V_0$, $T_C = 800$ K

Along CD; pressure is constant

$V \propto T$

At point D; P_0, V_0, $T_D = 100K$

$W_{AB} = \frac{-1}{2}(P_0 + 2P_0) \times 4V_0 = -6P_0V_0$

$W_{BC} = 0$

$W_{CD} = +4P_0(8V_0 - V_0) = 28P_0V_0$

$W_{DA} = \frac{-1}{2}[4P_0 + P_0] \times 3V_0 = -7.5P_0V_0$

Net work done

$W_{ABCDA} = -6P_0V_0 + 28P_0V_0 - 7.5P_0V_0$

$= +14.5\ P_0V_0$

At point A;

$P_04V_0 = nR(100K)$

$P_0V_0 = 25nR$

$|W_{ABCDA}| = 14.5 \times 25 \times 0.06 \times \frac{25}{3}$

$= 181.25$ J

6. **Ans (8.75)**

$\Delta H_{CD} = nC_{p,m}\ \Delta T$

$= 0.06 \times \frac{5}{2} \times \frac{25}{3}(100 - 800)$

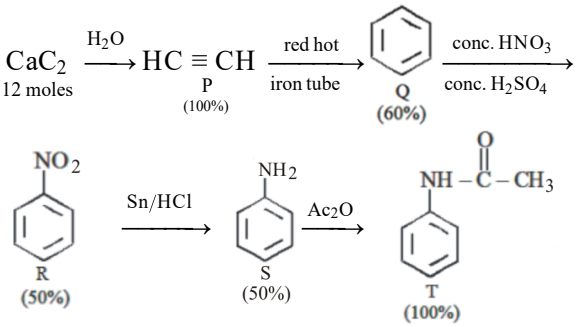
$= -875J = -8.75$ Lbar

$|\Delta H_{CD}| = 8.75$Lbar

PART-2 : CHEMISTRY

SECTION-II (ii)

7. **Ans (81)**



Actual yield of P = 12 moles

3 moles of P form = 1 mole of Q

So actual yield of Q = 60% of 4 = 2.4 moles

Actual yield of R = 1.2 moles

Actual yield of S = 0.6 moles

Actual yield of T = 0.6 moles

Molar mass of T = 135 g/mol

Mass of T = $135 \times 0.6 = 81$ g

8. **Ans (10)**

A sample can emit 10 different spectral lines when

$n_2 = 5$ and $n_1 = 1$

maximum energy emitted is 117.5 eV

$\Delta E = 13.6z^2\left(\frac{1}{1} - \frac{1}{25}\right) = 117.5eV \Rightarrow z = 3$

Photon of minimum energy is emitted from $5 \longrightarrow 4$ transition.

$\Delta E = 13.6 \times 9 \times \left(\frac{1}{16} - \frac{1}{25}\right)$

$= \frac{13.6 \times 9 \times 9}{16 \times 25} = 2.754eV$

Work function of metal = 1.254eV

$K.E._{max.}$ of photo electrons = 1.5eV

de-Broglie wavelength of photo electron is

$\lambda = \sqrt{\frac{150}{1.5}} = \sqrt{100}A^\circ$

$\lambda = 10A^\circ$

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9. **Ans (84)**

$H_2O \rightarrow H_2 + \frac{1}{2}O_2$ (nf = 4)

Moles of O_2 produced = $\frac{40F \times 0.75}{F \times 4}$ = 7.5 moles

$C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(\ell)$

moles $\frac{3.5}{28} = \frac{1}{8}$ 7.5

The heat released by $\frac{1}{8}$ moles of C_2H_4 is

$= C.\Delta T$

$= 3 \times 0.5 = 1.5KJ$

$\Delta U(KJ/mol) = \frac{-1.5 \times 28}{3.5}$ = -12KJ/mol

$C_2H_4(g) + 3O_2(g) \rightarrow 2CO_2(g) + 2H_2O(\ell)$

$\Delta H = \Delta U + \Delta ngRT$ (for one mole)

$= -12 + \frac{(-2) \times 8 \times 300}{1000}$

= -12-4.8 = -16.8 KJ/mol.

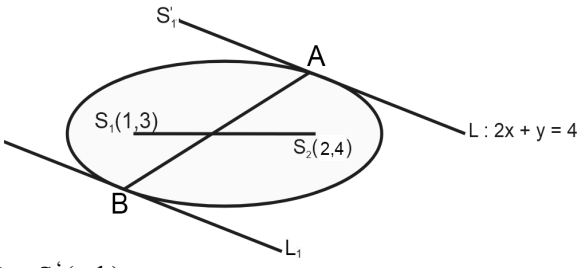
for 5 moles of ethene

$|\Delta H| = +16.8 \times 5 = 84KJ$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B)**



Let $S_1'(a, b)$

$\frac{a-1}{2} = \frac{b-3}{1} = \frac{-2(2+3-4)}{4+1}$

$\Rightarrow a = \frac{1}{5}, b = \frac{1}{5}$

$\Rightarrow S_1'\left(\frac{1}{5}, \frac{13}{5}\right)$

Now $S_1'S_2$: $y - 4 = \frac{\frac{13}{5} - 4}{\frac{1}{5} - 2}(x - 2)$

$\therefore$ 7x - 9y = -22 ---------- (1)

Also L : 2x + y = 4 ------- (2)

By (1) and (2) : A $\left(\frac{14}{25}, \frac{72}{25}\right)$

By (1) and (2) : A $\left(\frac{14}{25}, \frac{72}{25}\right)$

Now O is mid point of A and B.

here O $\left(\frac{3}{2}, \frac{7}{2}\right)$

Let B(α, β)

$\frac{\alpha + \frac{14}{25}}{2} = \frac{3}{2} \Rightarrow \alpha = \frac{61}{25}$

$\frac{\beta + \frac{72}{25}}{2} = \frac{7}{2} \Rightarrow \beta = \frac{103}{25}$

$\therefore$ B $\left(\frac{61}{25}, \frac{103}{25}\right)$

2. **Ans (B)**

$I = \int_1^2 (2^x + 2^{-x})\, d(2^x)$

$= \int_0^1 (2^x + 2^{-x})\, 2^x \ell n2 dx$

$= \ell n2 \int_0^1 (4^x + 1)\, dx$

$= \ell n2 \left[\frac{4^x}{\ell n4} + x\right]_0^1$

$= \ell n2 \left[\frac{4}{\ell n4} + 1 - \frac{1}{\ell n4}\right]$

$= \ell n2 \left[\frac{3}{\ell n4} + 1\right]$

$= \frac{3}{2} + \ell n2$

3. **Ans (C)**

Given limit

$= \lim_{x \to 0} \left(\frac{e^{\sin x} - e^x}{\sin x - x}\right)\left(\frac{x}{\sin x}\right)$

= 1

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4. **Ans (B)**

$$\sum_{i=1}^{20}(x_i-\bar{x})(y_i-\bar{y})=90$$

$$\Rightarrow \sum x_iy_i-\bar{x}\sum y_i-\bar{y}\sum x_i+\bar{x}\bar{y}\sum 1=90$$

$$\Rightarrow \sum x_iy_i-20\bar{x}\bar{y}=90$$

$$\Rightarrow \sigma^2=\frac{\sum_{i=1}^{20}(x_i-y_i)^2}{20}-(\bar{x}-\bar{y})^2$$

$$=\frac{\sum x_i^2}{20}+\frac{\sum y_i^2}{20}-\frac{2\sum x_iy_i}{20}-(\bar{x})^2-(\bar{y})^2+2\bar{x}\bar{y}$$

$$=16+9-\frac{2}{20}(90)=16$$

$$\Rightarrow \sigma=4$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (A,C)**

$$I=\int\frac{3x^4-2\left(x^4-16\right)}{x^8\left(x^4-16\right)}dx$$

$$=3\int\frac{1}{x^4\left(x^4-16\right)}dx-2\int\frac{1}{x^8}dx$$

$$=\frac{3}{16}\int\frac{x^4-\left(x^4-16\right)}{x^4\left(x^4-16\right)}dx+\frac{2}{7}\cdot\frac{1}{x^7}$$

$$=\frac{3}{16}\int\frac{1}{x^4-16}dx-\frac{3}{16}\int\frac{1}{x^4}dx+\frac{2}{7x^7}$$

$$=\frac{1}{8}\times\frac{3}{16}\int\frac{\left(x^2+4\right)-\left(x^2-4\right)}{\left(x^2+4\right)\left(x^2-4\right)}dx+\frac{3}{16}\cdot\frac{1}{3x^3}+\frac{2}{7x^7}$$

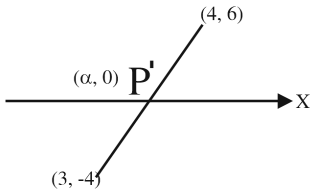
$$=\frac{3}{128}\int\left(\frac{1}{x^2-4}-\frac{1}{x^2+4}\right)dx+\frac{1}{16x^3}+\frac{2}{7x^7}$$

$$=\frac{3}{128}\left(\frac{1}{2.2}\ell n\left|\frac{x-2}{x+2}\right|-\frac{1}{2}\tan^{-1}\frac{x}{2}\right)+\frac{1}{16x^3}+\frac{2}{7x^7}+C$$

$$=\frac{3}{512}\ell n\left|\frac{x-2}{x+2}\right|-\frac{3}{256}\tan^{-1}\frac{x}{2}+\frac{1}{16x^3}+\frac{2}{7x^7}+C$$

6. **Ans (B,C,D)**

$$PA+PB=\sqrt{(\alpha-3)^2+16}+\sqrt{(\alpha-4)^2+36}$$



which is equivalent to P'A' + P'B' where P'(α, 0), A'(3, –4) & B'(4, 6)

it will be minimum when

$$\frac{4}{\alpha-3}=\frac{10}{1}\Rightarrow 2=5\alpha-15$$

$$\Rightarrow \alpha=\frac{17}{5}$$

So $P\left(\frac{17}{5},0,0\right)$

let Q (0, λ, 0)

Now $QA+QB=\sqrt{\lambda^2+25}+\sqrt{(\lambda-6)^2+16}$

it will be minimum when $\lambda=\frac{10}{3}$

$\therefore Q\left(0,\frac{10}{3},0\right)$

Area of ΔOPQ

$$=\frac{1}{2}\left|\overrightarrow{OP}\times\overrightarrow{OQ}\right|$$

$$=\frac{1}{2}\left|\frac{17}{5}\hat{i}\times\frac{10}{3}\hat{j}\right|$$

$$=\frac{17}{3}$$

Volume of tetrahedron OPAB

$$=\frac{1}{6}\left|\left[\overrightarrow{OP}\ \overrightarrow{OA}\ \overrightarrow{OB}\right]\right|$$

$$=\frac{1}{6}\begin{vmatrix}\frac{17}{5}&0&0\\3&0&4\\4&6&0\end{vmatrix}=\frac{68}{5}$$

equation of plane passing there P, Q, B :

$$\begin{vmatrix}x-\frac{17}{5}&y&z\\\frac{-17}{5}&\frac{10}{3}&0\\\frac{3}{5}&6&0\end{vmatrix}=0$$

$$\Rightarrow Z=0$$

distance of A from above plane = 4

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7. Ans (B,C,D)

$(x^2 + x + 1)[(f(x)-2)(x^2 - x + 1) - (f(x)+2)$

$(x^2+x+1)]=0$

$f(x)(x^2 - x + 1 - x^2 - x -1)-2x^2+2x - 2-2x^2-2x - 2 = 0$

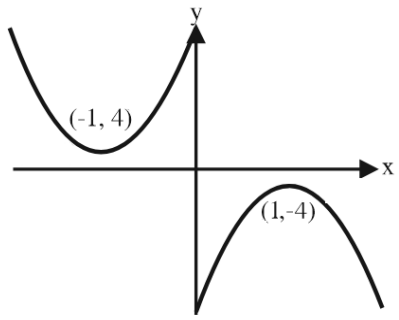
$f(x)(-2x) = 4x^2 + 4$

$f(x) = -2\left(x + \frac{1}{x}\right)$

$x + \frac{1}{x} \in (-\infty, -2] \cup [2, \infty)$

$f(x) \in (-\infty, -4] \cup [4, \infty)$

$f'(x) = -2\left(1 - \frac{1}{x^2}\right) = \frac{-2}{x^2}(x + 1)(x - 1)$



8. Ans (B,C)

The ways in which 4 runners can reach finishing line are as following :

(i) 1+1+1+1 $\Rightarrow \lfloor 4 = 24$ ways

(ii) 1+1+2 $\Rightarrow \frac{\lfloor 4}{\lfloor 1 \lfloor 1 \lfloor 2 \lfloor 2} \times \lfloor 3 = 36$ ways

(iii) 1 +3 $\Rightarrow \frac{\lfloor 4}{\lfloor 1 \lfloor 3} \times \lfloor 2 = 8$ ways

(iv) 2 + 2 $\Rightarrow \frac{\lfloor 4}{\lfloor 2 \lfloor 2 \lfloor 2} \times \lfloor 2 = 6$ ways

$\therefore P_4 = 24 + 36 + 8 + 6 + 1 = 75$ ways

The ways in which 5 runners can reach finishing line are as following :

(i) 1 + 1 + 1 + 1 + 1 $\Rightarrow \lfloor 5 = 120$ ways

(ii) 1+1+1+2 $\Rightarrow \frac{\lfloor 5}{\lfloor 1 \lfloor 1 \lfloor 1 \lfloor 2 \lfloor 3} \times \lfloor 4 = 240$ ways

(iii) 1+1+3 $\Rightarrow \frac{\lfloor 5}{\lfloor 1 \lfloor 1 \lfloor 2 \lfloor 3} \times \lfloor 3 = 60$ ways

(iv) 1+2+2 $\Rightarrow \frac{\lfloor 5}{\lfloor 1 \lfloor 2 \lfloor 2 \lfloor 2} \times \lfloor 3 = 90$ ways

(v) 1+4 $\Rightarrow \frac{\lfloor 5}{\lfloor 1 \lfloor 4} \times \lfloor 2 = 10$ ways

(vi) 2+3 $\Rightarrow \frac{\lfloor 5}{\lfloor 2 \lfloor 3} \times \lfloor 2 = 20$ ways

(vii) 5 $\Rightarrow$ 1 ways

$\therefore P_5 = 120 + 240 + 60 + 90 + 10 + 20 + 1 = 541$ ways

9. Ans (A,C)

$8x^2-14x + 3 = 0$

$\Rightarrow x = \frac{1}{4}, \frac{3}{2}$

$\Rightarrow \alpha = \frac{1}{4}, \beta = \frac{3}{2}$

Now

$\left(2\alpha + \sqrt{\frac{\beta}{2}}i\right)^{50} = \left(\frac{1}{2} + \frac{\sqrt{3}}{2}i\right)^{50} = (-w^2)^{50} = w$

&

$\left(2\alpha - \sqrt{\frac{\beta}{2}}i\right)^{50} = \left(\frac{1}{2} - \frac{\sqrt{3}}{2}i\right)^{50} = (-w)^{50} = w^2$

10. Ans (A,C,D)

$\ell nx \le \lambda x^3 \Rightarrow \lambda \geqslant \frac{\ell nx}{x^3}$

let $f(x) = \frac{\ell nx}{x^3}$

$f'(x) = \frac{x^2 - 3x^2\ell nx}{x^6}$

$= \frac{1 - 3\ell nx}{x^4}$

Maximum value of f(x)

$= f(e^{1/3}) = \frac{\ell n\left(e^{1/3}\right)}{e} = \frac{1}{3e}$

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PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (7.00)

$$D = \begin{vmatrix} 10 & 11 & 12 \\ 13 & 14 & 15 \\ 16 & 17 & 18 \end{vmatrix} = 0$$

$$D_1 = \begin{vmatrix} a & 11 & 12 \\ b & 14 & 15 \\ c-3 & 17 & 18 \end{vmatrix} = \begin{vmatrix} a & 11 & 1 \\ b & 14 & 1 \\ c-3 & 17 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} a & 11 & 1 \\ b-a & 3 & 0 \\ c-b-3 & 3 & 0 \end{vmatrix}$$

$\Rightarrow 3b - 3a - 3c + 3b + 9$

$\Rightarrow 6b - 3a - 3c + 9$

$\Rightarrow 3[2b - a - c + 3]$

$$D_2 = \begin{vmatrix} 10 & a & 12 \\ 13 & b & 15 \\ 16 & c-3 & 18 \end{vmatrix} = \begin{vmatrix} 10 & a & 12 \\ 3 & b-a & 3 \\ 3 & c-b-3 & 3 \end{vmatrix}$$

$$= \begin{vmatrix} 10 & a & 12 \\ 3 & b-a & 3 \\ 0 & c-2b+a-3 & 0 \end{vmatrix}$$

$= -(c - 2b + a - 3)[-6] \Rightarrow 6(c + a - 2b - 3)$

$$D_3 = \begin{vmatrix} 10 & 11 & a \\ 13 & 14 & b \\ 16 & 17 & c-3 \end{vmatrix} = \begin{vmatrix} 10 & 1 & a \\ 13 & 1 & b \\ 16 & 1 & c-3 \end{vmatrix}$$

$$= \begin{vmatrix} 10 & 1 & a \\ 3 & 0 & b-a \\ 3 & 0 & c-b-3 \end{vmatrix}$$

$= -1[3c - 3b - 9 - 3b + 3a]$

$= [6b - 3c - 3a + 9]$

$= 3[2b + 3 - a - c]$

system will be consistent if $D = D_1 = D_2 = D_3 = 0$

$\Rightarrow 2b + 3 - a - c = 0$

Now $|X| = a(-1) - 2(-b) + 1(-c)$

$= -a + 2b - c$

$= -3$

Now plane P : $-x + 2y - z + 3 = 0$

2. Ans (24.00)

$$D = \begin{vmatrix} 10 & 11 & 12 \\ 13 & 14 & 15 \\ 16 & 17 & 18 \end{vmatrix} = 0$$

$$D_1 = \begin{vmatrix} a & 11 & 12 \\ b & 14 & 15 \\ c-3 & 17 & 18 \end{vmatrix} = \begin{vmatrix} a & 11 & 1 \\ b & 14 & 1 \\ c-3 & 17 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} a & 11 & 1 \\ b-a & 3 & 0 \\ c-b-3 & 3 & 0 \end{vmatrix}$$

$\Rightarrow 3b - 3a - 3c + 3b + 9$

$\Rightarrow 6b - 3a - 3c + 9$

$\Rightarrow 3[2b - a - c + 3]$

$$D_2 = \begin{vmatrix} 10 & a & 12 \\ 13 & b & 15 \\ 16 & c-3 & 18 \end{vmatrix} = \begin{vmatrix} 10 & a & 12 \\ 3 & b-a & 3 \\ 3 & c-b-3 & 3 \end{vmatrix}$$

$$= \begin{vmatrix} 10 & a & 12 \\ 3 & b-a & 3 \\ 0 & c-2b+a-3 & 0 \end{vmatrix}$$

$= -(c - 2b + a - 3)[-6] \Rightarrow 6(c + a - 2b - 3)$

$$D_3 = \begin{vmatrix} 10 & 11 & a \\ 13 & 14 & b \\ 16 & 17 & c-3 \end{vmatrix} = \begin{vmatrix} 10 & 1 & a \\ 13 & 1 & b \\ 16 & 1 & c-3 \end{vmatrix}$$

$$= \begin{vmatrix} 10 & 1 & a \\ 3 & 0 & b-a \\ 3 & 0 & c-b-3 \end{vmatrix}$$

$= -1[3c - 3b - 9 - 3b + 3a]$

$= [6b - 3c - 3a + 9]$

$= 3[2b + 3 - a - c]$

system will be consistent if $D = D_1 = D_2 = D_3 = 0$

$\Rightarrow 2b + 3 - a - c = 0$

Now $|X| = a(-1) - 2(-b) + 1(-c)$

$= -a + 2b - c$

$= -3$

Now plane P : $-x + 2y - z + 3 = 0$

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3. Ans (26.00)

$f(x)f(y) + 2 = f(x) + f(y) + f(xy)$ ....(1)

putting $x = y = 1$

$f^2(1) + 2 = 3f(1)$

$f^2(1) - 3f(1) + 2 = 0$

$f(1) = 1, 2$

since $f(x)$ is one-one & $f(0) = 1$

so $f(1) = 2$

replace y with $\frac{1}{x}$ in (1)

$f(x)f\left(\frac{1}{x}\right) + 2 = f(x) + f\left(\frac{1}{x}\right) + 2$

$f(x) = 1 \pm x^n$

but $f(1) = 2$ & $f'(1) = 2$

$\therefore f(x) = 1 + x^2$

4. Ans (3.00)

$f(x)f(y) + 2 = f(x) + f(y) + f(xy)$ ....(1)

putting $x = y = 1$

$f^2(1) + 2 = 3f(1)$

$f^2(1) - 3f(1) + 2 = 0$

$f(1) = 1, 2$

since $f(x)$ is one-one & $f(0) = 1$

so $f(1) = 2$

replace y with $\frac{1}{x}$ in (1)

$f(x)f\left(\frac{1}{x}\right) + 2 = f(x) + f\left(\frac{1}{x}\right) + 2$

$f(x) = 1 \pm x^n$

but $f(1) = 2$ & $f'(1) = 2$

$\therefore f(x) = 1 + x^2$

5. Ans (3.00)

When $n = 1$, $a_1 = 5S_1 + 1 = 5a_1 + 1$

$\Rightarrow a_1 = \frac{-1}{4}$

for $n \geqslant 1$

$a_{n+1} - a_n = 5S_{n+1} - 5S_n$

$\Rightarrow a_{n+1} - a_n = 5a_{n+1}$

$\Rightarrow 4a_{n+1} = -a_n \Rightarrow \frac{a_{n+1}}{a_n} = \frac{-1}{4}$

there fore $\{a_n\}$ is a GP with $a_1 = \frac{-1}{4}$ & common ratio $r = \frac{-1}{4}$

So $a_{10} = \left(\frac{-1}{4}\right)\left(\frac{-1}{4}\right)^9 = \frac{1}{4^{10}}$

Also $a_{12} = \left(\frac{-1}{4}\right)\left(\frac{-1}{4}\right)^{11} = \frac{1}{4^{12}}$

So $b_{12} = \frac{4 + a_{12}}{1 - a_{12}} \Rightarrow \frac{4 + \frac{1}{4^{12}}}{1 - \frac{1}{4^{12}}} \Rightarrow \frac{4^{13} + 1}{4^{12} - 1}$

6. Ans (0.50)

When $n = 1$, $a_1 = 5S_1 + 1 = 5a_1 + 1$

$\Rightarrow a_1 = \frac{-1}{4}$

for $n \geqslant 1$

$a_{n+1} - a_n = 5S_{n+1} - 5S_n$

$\Rightarrow a_{n+1} - a_n = 5a_{n+1}$

$\Rightarrow 4a_{n+1} = -a_n \Rightarrow \frac{a_{n+1}}{a_n} = \frac{-1}{4}$

there fore $\{a_n\}$ is a GP with $a_1 = \frac{-1}{4}$ & common ratio $r = \frac{-1}{4}$

So $a_{10} = \left(\frac{-1}{4}\right)\left(\frac{-1}{4}\right)^9 = \frac{1}{4^{10}}$

Also $a_{12} = \left(\frac{-1}{4}\right)\left(\frac{-1}{4}\right)^{11} = \frac{1}{4^{12}}$

So $b_{12} = \frac{4 + a_{12}}{1 - a_{12}} \Rightarrow \frac{4 + \frac{1}{4^{12}}}{1 - \frac{1}{4^{12}}} \Rightarrow \frac{4^{13} + 1}{4^{12} - 1}$

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PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (32)

Let P_k denotes the probability that code A is used in the kth week.

$P_{k+1} = \frac{1}{4}(1 - P_k)$

$P_{k+1} - \frac{1}{5} = \frac{-1}{4}\left(P_k - \frac{1}{5}\right)$

as $P_1 = 1$, $\left\{P_K - \frac{1}{5}\right\}$ is a GP with $\frac{4}{5}$ as the first term & $\left(\frac{-1}{4}\right)$ as the common ratio.

So $P_k - \frac{1}{5} = \frac{4}{5}\left(-\frac{1}{4}\right)^{k-1}$

So $P_9 = \frac{4}{5}\left(-\frac{1}{4}\right)^8 + \frac{1}{5} = \frac{3273}{16384}$

8. Ans (4)

$z = e^{i\theta}$

let $y = |z^3 - z + \alpha|$

$y = |(\cos 3\theta + i \sin 3\theta) - (\cos\theta + i\sin\theta) + 2|$

$y^2 = (\cos 3\theta - \cos\theta + 2)^2 + (\sin 3\theta - \sin\theta)^2$

$y^2 = 4\cos 3\theta - 2\cos 2\theta - 4\cos\theta + 6$

Let $\cos\theta = x$

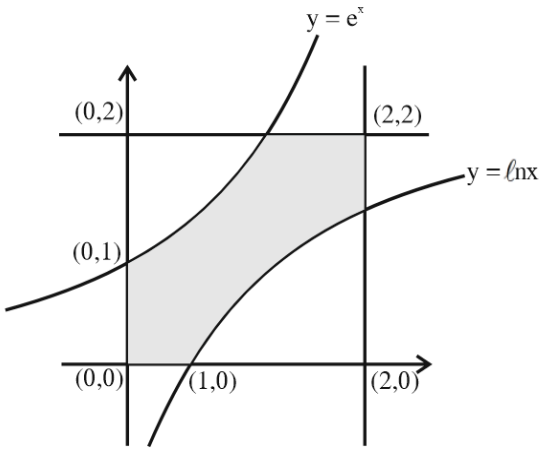
$y^2 = 16x^3 - 4x^2 - 16x + 8$

so $x \in [-1, 1]$

$y^2_{max} = 13$

$y_{max} = \sqrt{13}$

9. Ans (5)



Area = $4 - 2\int_1^2 \ell n x dx$

$= 6 - 4\ell n 2$

$a = 6, b = 4$

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